

MACROSCALE BRAIN STRUCTURAL NETWORK COUPLING IS RELATED TO AD PROGRESSION

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ABSTRACT

Morphological Similarity Network (MSN) and Structural Connectivity Network (SCN) represent the two primary brain network models extensively employed for investigating structural connectivity disruptions in Alzheimer's disease (AD). However, the spatial relationship between MSN and SCN and its connection to AD progression remains unclear. This study explored the association between the coupling of MSN and SCN and AD progression at local and global levels via a multisite, multimodal MRI study (N=721). Our results demonstrate a significant reduction in global and local coupling with increasing cognitive decline from mild cognitive impairment to AD dementia. Furthermore, the coupling of SCN and MSN is linked to cognitive function, i.e., MMSE score. Genes associated with coupling alterations in AD are notably enriched in pathways related to the regulation of monoatomic ion transport. Our findings contribute novel insights into understanding the multimodal brain connectome dysfunction in AD.

Index Terms— *Alzheimer's disease, multimodal brain network, coupling, gene expression*

1. INTRODUCTION

Alzheimer's disease (AD) stands as a prevailing neurodegenerative affliction known for its hallmark manifestation of cognitive deterioration [1]. Extensive prior investigations have unveiled a prevailing theme: the impaired functionality of structural connections emerges as a recurring neuroimaging metric within the context of AD. Specifically, researchers have traditionally harnessed two principal approaches to assess these structural connections: the Diffusion Weighted Imaging (DWI)-based Structural Connectivity Network (SCN) and the T1-weighted (T1)-based Morphological Similarity Network (MSN) [2].

However, the intricate relationship between SCN and MSN is not clear yet. On the one hand, the spatial similarity of the SCN and MSN is uncertain. On the other hand, the relationship between the coupling of SCN and MSN and AD progression is not studied well yet [3, 4]. Investigating the coupling of these two frequently employed brain networks is

crucial for elucidating the cognitive and biological foundations of distinct brain models and gaining a deeper understanding of the underlying AD mechanisms.

This study introduced a novel approach via the spatial similarity between the SCN and a novel MSN, regional radiomics similarity network (R2SN) [5-7], to obtain the coupling of a multimodal brain network at the local and global levels. Then, we quantified the alterations in the AD group via a meta-analysis framework. At last, we elucidated the underlying mechanisms behind these alterations of the coupling between R2SN and SCN in AD via gene enrichment analysis.

2. MATERIALS AND METHODS

2.1 Data information and preprocessing

This study comprised 721 participants (Table 1), including individuals with 278 ADs, 188 mild cognitive impairments (MCI), and 255 normal controls (NC), as well as their paired T1 images, DWI images, demographic and psychological information collected from seven sites participating in the MCADI (Multi-Center Alzheimer's Disease Imaging) [8].

The T1 image was aligned to the Montreal Neurological Institute (MNI) space using Advanced Normalization Tools (ANTs) (<https://github.com/ANTsX/ANTsPy>). Then, the individual R2SN was constructed via an open-access toolkit (<https://github.com/YongLiuLab/>). Briefly, the nodes of the R2SN represent the brain regions defined by the Brainnetome atlas [9], and the edges of the R2SN are defined based on Pearson's correlations between normalized interregional radiomics features (N = 25) [5].

DWI data preprocessing comprised skull dissection, eddy current correction, registration, and normalization. After that, the SCN based on the number of fiber connections between brain regions of the Brainnetome Atlas was constructed via deterministic fiber tracking.

2.2 Coupling between R2SN and SCN

To evaluate the relationship between R2SN and SCN, we quantified the coupling between the two networks at both global and local levels. For the global coupling score, we calculated Pearson's correlations between all edges of the

R2SN and SCN with corresponding vectors, which were all edges of the network, each of size 30135×1 (representing the upper triangular matrix of 246×246).

$Global\ coupling\ score = Corr([M_{ij} \in R2SN], [N_{ij} \in SCN])$

To capture the spatial similarity for each node, we further calculated Pearson's correlations between the connections associated with each node in the R2SN and SCN for the local coupling score.

$Local\ coupling\ score_i = Corr([x_{ij}|_{j=1..246} \in R2SN], [y_{ij}|_{j=1..246} \in SCN])$ where i and j represent the *i*th and *j*th brain regions, respectively. **Fig. 1** shows the overview of the coupling pipeline.

Table 1. Demographics and psychological statistics of the included subjects.

	Groups	n	Age	P	Gender (M/F)	P	MMSE	P
S01	NC	41	68.6±6.7	0.764	19/22	0.747	28.5±1.4	<0.001
	MCI	34	69.5±8.8		13/21		26.6±2.5	
	AD	44	69.9±8.8		20/24		17.3±6.5	
S02	NC	21	68.6±4.7	0.054	11/10	0.096	28.9±1.1	<0.001
	MCI	21	74.1±8.0		11/10		26.8±1.7	
	AD	24	72.8±8.3		6/18		19.2±4.6	
S03	NC	24	65.5±6.2	0.409	9/15	0.286	28.8±1.2	<0.001
	MCI	33	65.4±8.3		10/23		25.9±2.5	
	AD	37	67.7±8.3		18/19		15.8±5.5	
S04	NC	42	65.5±6.8	0.244	12/30	0.209	28.4±1.8	<0.001
	MCI	16	66.1±7.4		8/8		24.8±1.5	
	AD	63	67.8±7.2		27/36		19.8±3.0	
S05	NC	64	66.6±6.3	0.236	25/39	0.755	28.1±2.2	<0.001
	MCI	29	69.1±9.5		13/16		22.9±3.1	
	AD	39	68.8±8.8		14/25		16.6±6.7	
S06	NC	21	65.0±8.2	0.081	7/14	0.368	28.5±1.4	<0.001
	MCI	18	70.2±7.9		10/8		21.9±5.0	
	AD	30	65.3±7.8		14/16		11.0±6.3	
S07	NC	42	68.3±8.0	0.504	20/22	0.508	28.8±1.2	<0.001
	MCI	37	69.8±6.9		13/24		26.3±2.6	
	AD	41	70.2±8.7		16/25		15.8±5.8	

χ^2 -tests tested gender comparisons between groups; one-way Analysis of Variance (ANOVA) was applied for age and MMSE score comparisons.

2.3 Difference analysis for local coupling between AD and NC

We employed a two-sample, two-sided t-test for NC and AD groups at each site to explore the difference in the local coupling in AD and NC. Subsequently, we utilized a meta-analysis approach to obtain a combined z-score from p values across the seven sites after controlling age and gender by the linear regression model. In particular, $z_i = \varphi^{-1}(1 - p_i/2)$, where φ^{-1} is the standard normal cumulative distribution function. The combined z-score was computed using the following Liptak-Stouffer formula [10]:

$$z = \sum_{i=1}^k w_i z_i / \sqrt{\sum_{i=1}^k w_i^2}$$

where w_i is the square root of the sample size of dataset *i*, and *k* is the number of datasets.

2.4 Associations between clinical scores and coupling

To explore the relationship between coupling and cognitive performance, we performed a Pearson's correlation analysis, investigating the correlations between global coupling scores and Mini-Mental State Examination (MMSE) scores within the AD and MCI groups. Within the AD and MCI groups, we also performed Pearson's correlation analyses to explore the associations between MMSE scores and local coupling scores for each region. We accounted for potential confounding factors, including age and gender effects.

2.5 Associations between gene expression and local coupling alterations

To explore the biological mechanism of the alterations of the local coupling score in AD, a gene enrichment analysis was performed in the present study. We obtained 238×15633 gene expression matrix via the Abagen toolbox (<https://abagen.readthedocs.io/en/stable/>) and projected the Allen Human Brain atlas (<http://human.brain-map.org/>) to the Brainnetome Atlas. Then, we performed a partial least squares (PLS) regression to investigate the association between the Z-map of the local coupling score in AD vs. NC and gene expression. The permutation analysis (*n* = 5,000) corrected for spatial autocorrelation of the PLS components. The PLS weight of each gene was transformed into a z-score value by dividing the weight by the standard deviation of the corresponding weights derived from 5,000 bootstrapping instances. Next, 15633 genes were ranked based on their corresponding weight value. Finally, we performed gene-set enrichment analysis based on the top 500 genes by the Metascape platform [11].

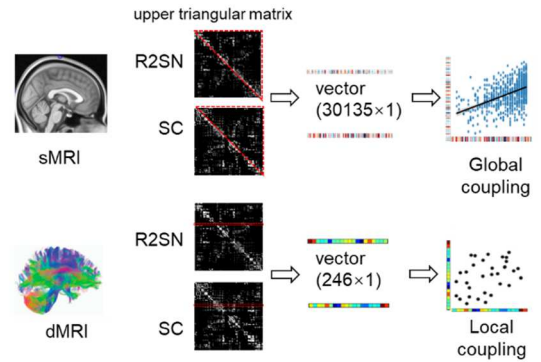


Fig.1 The schematic diagram of the coupling construction.

3. RESULT

3.1 Demographic and clinical characteristics

Among the NC, MCI, and AD groups, there were no significant differences in the age and sex ratio of the subjects. The MMSE score significantly differed among the NC, MCI, and AD groups with *p*<0.001 (Table 1).

3.2 Global coupling patterns are related to cognitively impaired states

The global coupling scores for NCs were 0.054 ± 0.013 , while the global coupling scores for MCI and AD patients were 0.050 ± 0.012 and 0.044 ± 0.014 , respectively (Fig 2A). Subsequent pairwise t-tests revealed that the global coupling scores in the AD group were significantly lower than those in the MCI group ($p < 0.001$) and the NC group ($p < 0.001$). Additionally, the global coupling scores in the MCI group were significantly lower than those in the NC group ($p < 0.001$). Furthermore, the MMSE scores exhibited positive correlations with the global coupling scores ($r = 0.28$, $p < 0.001$, Fig 2B).

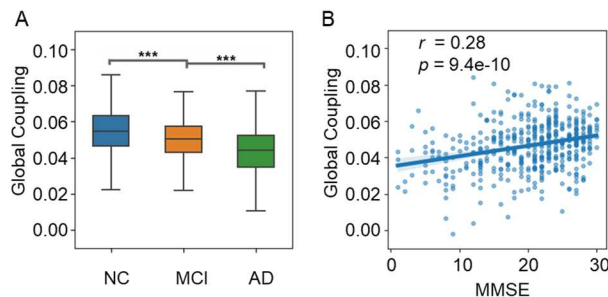


Fig 2. (A) The global coupling scores significantly differed among the NC, MCI, and AD groups (***, $p < 0.001$). (B) The correlation between global coupling scores and MMSE scores within MCI and AD groups.

3.3 Local coupling scores suggested regional heterogeneity and varied in the AD continuum

After elucidating the global coupling patterns, we leveraged the neuroimaging data to delve into regional heterogeneity. In general, among the AD, MCI, and NC groups, the distribution of regional R2SN coupling was broadly similar (Fig 3A). We further compared regional coupling across the NC and AD groups with age and sex as covariates and observed a primarily decreasing pattern from the NC to AD groups (Fig 3B). The local coupling scores were significantly decreased in the AD group, mainly including the superior frontal gyrus, middle frontal gyrus, middle temporal gyrus, and cingulate gyrus. In contrast, the AD group showed significantly higher local coupling scores in the thalamus, basal ganglia, amygdala, and parahippocampal gyrus ($p < 0.05$, Bonferroni corrected for $N = 246$).

3.4 Coupling scores associated with clinical scores in the AD and MCI

Forty-three regional local coupling scores were significantly correlated with the MMSE scores in the MCI and AD groups (Fig 3C). Notable positive correlations emerged in the superior frontal gyrus, middle frontal gyrus, and middle temporal gyrus, while significant negative correlations were

observed in the thalamus, basal ganglia, and parahippocampal gyrus ($p < 0.05$, FDR corrected).

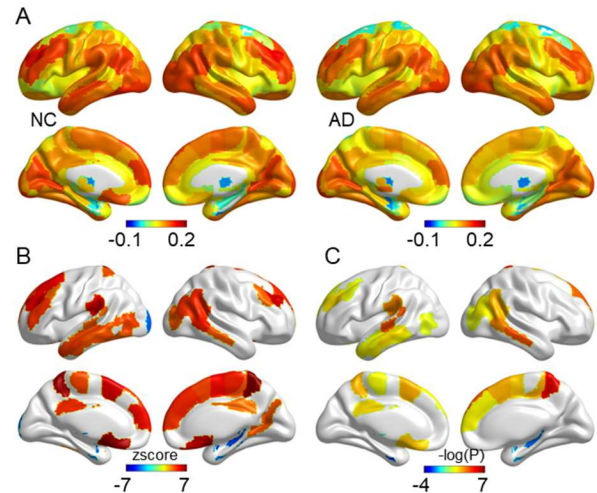


Fig 3. (A) The local coupling pattern of NC and AD. (B) Statistically significant local coupling differences between NC and AD. (C) The correlation map between local coupling and MMSE scores in the AD and MCI groups.

3.5 Associations between coupling differences and gene expression

The first PLS component (PLS1) accounted for a 16% variance in gene expression and significantly correlated with the group difference of local coupling scores ($r = 0.42$, $p = 1.8e-11$, corrected for the Spin test) (Fig 4A and Fig 4B). We found the Neuronal System (R-HSA-112316, $p_{FDR} = 8.9e-13$), regulation of monoatomic ion transport (GO:0043269, $p_{FDR} = 4.5e-10$), synaptic signaling (GO:0099536, $p_{FDR} = 1.4e-8$), and modulation of chemical synaptic transmission (GO:0050804, $p_{FDR} = 5.2e-8$) are the most significantly correlated biological process for the alterations of the coupling (Fig 4C).

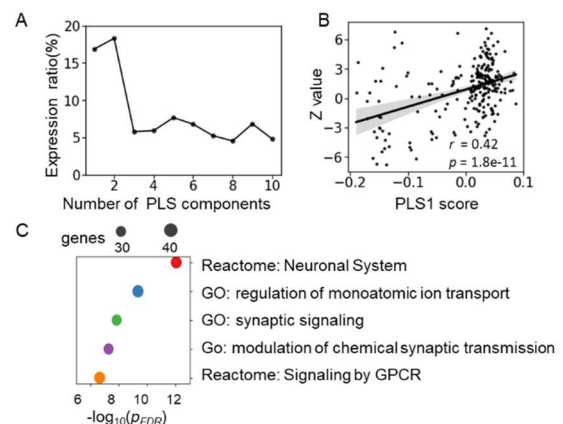


Fig 4. (A) The first ten components of the PLS regression analysis. (B) Spatial association between AD vs. NC differences and PLS1 scores. (C) Gene enrichment for PLS1 genes.

4. DISCUSSION

We calculated both global and local coupling to explore the changes that occur in AD. Global and local coupling consistently declined as individuals progressed from normal cognitive function to AD. We found a strong correlation between global coupling and the cognitive state of patients, suggesting that global coupling can indicate the disease course in AD. We hypothesized that lower coupling values may signify the involvement of more abnormal pathophysiological mechanisms, which could contribute to the poor prognosis of AD. This hypothesis is also supported by the progressive disease outcomes observed in individuals with MCI.

Regarding local coupling patterns, we found higher coupling in the frontal and temporal lobes, indicating that underlying solid physical connections in those regions support morphological similarity. In contrast, the subcortical regions exhibited lower coupling, potentially due to the substantial structural differences between the subcortical and cortical regions. Significant local differences between NC and AD were primarily found in the default mode network (DMN). The DMN is responsible for self-cognition, mind, memory, emotion, and other essential functions, and changes in DMN are a vital characteristic of AD [12]. These observations provide valuable insights into the structural and morphology changes in the brain associated with AD.

5. LIMITATION AND FUTURE DIRECTION

The age gap between our data and AHBA data could cause potentially flawed results in the radiogenomics analysis. Future research could further explore the underlying molecular mechanisms by integrating gene expression data from participants with AD to evaluate the molecular basis for these coupling patterns.

6. CONCLUSION

We have introduced an innovative approach to exploring the intricate correlation between the SCN and the MSN and have extended this investigation to uncover their connection to AD progression. This study presents a novel method and fresh insights for comprehending the relationship between brain morphology abnormalities and brain structural connective disconnection and its biological basis in AD.

7. COMPLIANCE WITH ETHICAL STANDARDS

This study was consistent with the Principles of the Declaration of Helsinki and approved by the Medical Research Ethics Committee and Institutional Review Board.

8. ACKNOWLEDGMENTS

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